1. Exercise Problems - Problems have been taken from Topologywithouttears by Sidney Morris

1.1. **Chapter 1.** Exercise 1.1

Problem 1

a) Is not a topology since $\{a, f\} \cap \{b, f\} = \{f\} \notin \mathcal{T}_1$.

b) Is not a topology since $\{a, b, f\} \cap \{a, b, d\} = \{a, b\} \notin \mathcal{T}_2$

c) Is a topology since the unison of each subset with respect to another subset and the intersection of each subset with respect to another subsest is contained in T_3.

Problem 2

a) Is not a topology since $c \cup b = b, c \notin T_1$

b) Is not a topology since $b, d, e \cap a, b, d = b, d \notin T_2$.

c) Is a topology since the unions of each subsset with respect to another subset and the intersection of each subset with respect to another subset is contained in T_3.

Problem 4 Let $(X, \mathcal{T})$ be any topological space. Verify that the **intersection of any finite number of members of** $\mathcal{T}$ is a member of $\mathcal{T}$.

Proof. Case: $n = 3$

Let $A, B, C \in \mathcal{T}$

Then $A \cap B \cap C = (A \cap B) \cap C$.

Since $A, B \in \mathcal{T}$, by definition part 1.3, $A \cap B \in \mathcal{T}$

Since $C \in \mathcal{T}$, $(A \cap B) \cap C \in \mathcal{T}$

Therefore, $A \cap B \cap C \in \mathcal{T}$

Case: $n = k$

Assume $n = k$ is true, i.e. the intersection of any k open sets in $\mathcal{T}$ is in $\mathcal{T}$

Let $O_1, O_2, ..., O_k \in \mathcal{T}$, then $O_1 \cap O_2 ... O_k \in \mathcal{T}$

Then $(O_1 \cap O_2 \cap ... \cap O_k) \cap O_(k + 1)$

Hence, the intersection of any finite number of members of T is a member of T. □

Problem 7

a)

$$\mathcal{T}_1 = \{X, \emptyset, \{a\}, \{b\}\}$$
$$\mathcal{T}_2 = \{X, \emptyset, \{a\}\}$$
$$\mathcal{T}_3 = \{X, \emptyset, ??\}$$

b) $(Y, T) = \{Y, \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

Problem 8

Proof. Let X be an infinite set and $\mathcal{T}$ a topology on X. Assume that every infinite subset of X is in $\mathcal{T}$.

Note that x is countably infinite, i.e. $x = \{x_1, x_2, x_3, ...x_i...\}$

Given that every infinite subset of X is in $\mathcal{T}$ then there are two sets of X, the odd set and the even set, i.e. $O = \{x_1, x_3, x_5...\}$ and $E = \{x_2, x_4, x_6...\}$

Let $x_1 \in X$

So, there is a set A and B such that $A = O \cup \{x_i\}$ and $B = E \cup \{x_i\}$.

Then, by the definition of Topology, $A \cap B = \{x_i\}$.

Now, $A \in \mathcal{T}$ and $B \in \mathcal{T}$.

By proposition 1.1.9, $A \cap B \in \mathcal{T}$.
Hence, $\mathcal{T}$ is the discrete topology.

□

Exercise 1.2
Problem 2 - Prove that (X,T) is a discrete space.

Proof. Let (X, T) be a topological space with the property that every subset is closed.
By definition 1.1.6, if X is a non-empty set and $\mathcal{T}$ is the collection of all subsets of X, then $\mathcal{T}$ is called the discrete topology. The topological space space (X, $\mathcal{T}$) is called a discrete space.
Since a topological space with closed subsets has non-empty sets, then $\mathcal{T}$ would be the discrete topology.
So, (X, $\mathcal{T}$) is a discrete space. □

Exercise 1.3
Problem 2 - Is the topology T described in Exercise 1.1 6 (ii) the finite closed topology?

Proof. Let $\mathbb{N}$ be the set of all positive integers. Assume that the topology $\mathcal{T}$ consists of $\mathbb{N}$, $\emptyset$, and every set $\{n, n+1, ...\}$ for any positive integer n.
Let $A \subseteq \mathbb{N}$ be a closed set under $\mathcal{T}$.
Then $A = O^c$ for some $O \in \mathcal{T}$.
Assume $o \neq \emptyset$, then $o = \{n, n+1, ...\}$.
Then $O^c = \{1, 2...n-1\} = A$ is finite. □

Problem 4 - Let $\mathcal{T}$ be the finite-closed topology on a set X. If $\mathcal{T}$ is also the discrete topology, prove that the set X is finite.

Proof. Given that $\mathcal{T}$ is finite closed and discrete then the singleton set $\{x\} \subset X$ are open in $\mathcal{T}$.
So, $\{x\}^c$ must be closed.
Note that $\{x\}^c = X \setminus \{x\}$ but $\mathcal{T}$ is finite.
Then $\{x\}^c$ is finite.
Suppose $|\{x\}^c| = n$ for some $n \in \mathbb{N}$.
Thus, $|X| = n + 1$, which is finite. □

1.2. **Chapter 2.** Exercise 2.1
Problem 1. Prove that if $a, b \in \mathbb{R}$ then neither [a,b) nor (a,b] is an open subset of $\mathbb{R}$. Also show that neither is a closed subset of $\mathbb{R}$

Proof. Let $a, b \in \mathbb{R}$. Assume that $a < b$.
By definition 2.1.1, there must be an element $x \in S$ defined by $(a, b \in \mathbb{R} : a < b$ such that $x \in (a, b) \subseteq S$
Note that [a,b) means that a is closed.
So [a,b) cannot be an open subset as it would contradict definition 2.1.1.
Also, (a,b] means that b is closed.
So (a,b] cannot be an open subset as it would also contradict definition 2.1.1.
Thus, neither [a,b) nor (a,b] is an open subset of $\mathbb{R}$.
Neither can be an open subset of $\mathbb{R}$ as it would also contradict definition 2.1.1.

□

Problem 2. Prove that the sets $[a, \infty)$ and $(-\infty, a]$ are closed subsets of $\mathbb{R}$.

Proof. By definition, a closed subset must have a complement that is open.

Let $(X, \mathcal{T})$ be a topological space. Let A be a subset of $(X, \mathcal{T})$. Then its complement, X A, is an open subset of $(X, \mathcal{T})$.

□

Exercise 2.2

Problem 3 - Let $\mathcal{B}$ be the collection of all open intervals (a, b) in $\mathbb{R}$ with $a < b$ and a and b rational numbers. Prove that $\mathcal{B}$ is a basis for the euclidean topology.

Proof. Let $(X, \mathcal{T})$ be a topological space. Let β be the collection of all open intervals (a, b) in $\mathbb{R}$ with $a < b$ and a and b are rational numbers.

By definition 2.2.2, a collection $\mathcal{B}$ of open subsets of X is said to be a basis if every open set is a union of members of β.

So, the union of members of β can be expressed as $\mathcal{T}_1 \cup \mathcal{T}_2 \cup ... \cup \mathcal{T}_n$.

Note that a euclidean topology has an $x \in S$ where $a, b \in \mathbb{R}$ such that $x \in (a, b) \subseteq S$. Thus, β is a basis for the euclidean topology.

□

Exercise 2.3

Problem 1.

i. Yes since an open square is a form of an open rectangle.

ii. Yes. Open rectangles can be replaced by open discs.

iii. Yes since an open square is a form of an open rectangle.

iv. Yes. See example 2.2.9

v. Yes. See example 2.3.5

Problem 3.

Proof. Let $\beta = \{(a, b] : a, b \in \mathbb{R}, a < b\}$.

Note that β is a basis for a topology $\mathcal{T}$ on $\mathbb{R}$ and $\mathcal{T}$ is not the euclidean topology on $\mathbb{R}$.

Observe that $(a, b]$ is an open set in $\mathbb{R}$ with topology $\mathcal{T}$

So, each interval (a, b) is open in $(\mathbb{R}, \mathcal{T})$

□

Problem 5.

i.

Proof. Let $(X, \mathcal{T})$ be a topological space. Assume that there is an open interval of the form (a, ∞).

So, there is a set S, where $\{x_1, x_2\} \cap \{x_2, x_3\} \cap \{x_n, x_(n+1)\} \subset S$ for some n.

Note that $S \subset (a, \infty)$

Since S is finite, then it is said to be a sub-basis.

Also, there exists $(a, \infty) \in \mathbb{R}$, with $a < \infty$ such that $x \in (a, \infty) \subset K$.

Hence, the collection of all open intervals of the form (a, ∞) form a sub-basis for the Euclidean topology on $\mathbb{R}$.

□

1.3. **Chapter 3.** Exercise 3.1

Problem 1.

a)

i. limit point($\{a\}$) = $\emptyset$

ii. limit point($\{b, c\}$) = b, d, e.

iii. limit point($\{a, c, d\}$) = b, c, d, e

iv. limit point($\{b, d, e\}$) = b, e

Note that the closed sets are $\{b, c, d, e\}, \{a, b, e\}, \{b, e\}$, and $\{a\}$.

b/c)

i. a

ii. b,c,d,e

iii. X

iv. b,c,d,e

Problem 2.

i.

Proof. Let S be a set in $(\mathbb{Z}, \tau)$ consisting of $\{1, 2, 3, ..., 10\}$. Then its complement is $\{11, 12, 13...\}$.

So, the union of anywhere from 2 to 10 singleton sets from n=1 to n=10 is the set of limit points, e.g. $\{1\} \cup \{2\}$ is one set consisting of two limit points: 1 and 2. □

ii.

The set E is infinite and thus not a finite closed topology. Note that its complement would be the set of all odd integers, O = 1,3,5,...n+1

Exercise 3.2

Problem 1.

Proof. Let A be a subset of the topological space (X, τ).

Assume that every neighbourhood of each point in $X \setminus A$ intersects A nontrivially.

So, $i_x \in X \setminus A$. Also, $i_x \cap A \neq \emptyset$

Note that $i_x \in U \subseteq N$.

Now, $\bar{A} = A \cup A'$

It suffices to show that $\bar{A} = i_x \cup A$

Hence, A is dense in X. □

Problem 2.

i.

Proof. Let A and B be subsets of a topological space (X, τ).

By definition 3.1.9, $A \bar{\cap} B = (A \cup A') \cap (B \cup B')$

So, there is a point $p \in A \bar{\cap} B$ such that $p \in A \cup A'$ and $p \in B \cup B'$.

Also, $\bar{A} \cap \bar{B} = (A \cup A') \cap (B \cup B')$.

So, there is a point $p \in \bar{A} \cap \bar{B}$, i.e. $p \in A \cup A'$ and $p \in B \cup B'$

Now, $p \in A \bar{\cap} B$ and $p \in \bar{A} \cap \bar{B}$

Thus, $A \bar{\cap} B \subseteq \bar{A} \cap \bar{B}$ □

Exercise 3.3

Problem 1.

a. Prove that the real number a is the infimum of S if and only if -a is the supremum of T.

Proof. Let S be a set of real numbers and $T = \{x : -x \in S\}$.
Case 1.
Assume that the real number a is the infimum of S.
Note that T can be rewritten as T = $\{a : -a \in \mathbb{R}\}$.
Since $-a \in \mathbb{R}$ and a is the infimum of S, then the supremum of T is -a.
Case 2.
Assume that -a is the supremum of T.
Since sup(T) = -a and $T = \{a : -a \in S\}$ then a is the infimum of S.
Hence, the real number a is the infimum of S and -a is the supremum of T.

□

Problem 4.

Proof. Let $X = \{a, b, c, d, e, f\}$ and $\tau_1 = \{\emptyset, X, \{a\}, \{c, d\}, \{a, c, d\}, \{b, c, d, e, f\}\}$
Note that the complements of $\mathcal{T}_1$ are $\{\emptyset, X, \{b, c, d, e, f\}, \{a, b, e, f\}, \{b, e, f\}, \{a\}\}$
Since $\{a\}$ and $\{b, c, d, e, f\}$ are clopen then the topology $\mathcal{T}_1$ is not connected as it fails to satisfy Definition 3.3.4, i.e. the only clopen sets allowed are X and $\emptyset$. □

1.4. **Chapter 4.** Exercise 4.1
Problem 1.
Note that the topology $\mathcal{T}_Y$ is the intersection of the topology $\mathcal{T}$ and the set Y, i.e. $\mathcal{T}_Y = \mathcal{T} \cap Y = \{\emptyset, \{a\}, \{a, c\}, \{a, e\}\}$.
In the same way, the topology $\mathcal{T}_Z$ is the intersection of the topology $\mathcal{T}$ and the set Z, i.e. $\mathcal{T}_Z = \mathcal{T} \cap Y = \{\emptyset\{c, d\}, \{b, c, d\}, \{b, e\}\}$
Problem 6.

Proof. Let $(X, \mathcal{T})$ be a discrete topology.
Assume that there is a nonempty set $Y \subset X$
Note that there is a point $p \in Y$ such that the intersection of Y and $\mathcal{T}$ will contain p

□

Problem 7.

Proof. Let $(X, \mathcal{T})$ be an indiscrete topology.
Assume that there is a nonempty set $Y \subset X$
Note that $\mathcal{T} \cap Y = \{Y, \emptyset\}$
So, $(Y, \mathcal{T})$ is an indiscrete subtopology of the original topology $(X, \mathcal{T})$.

□

Exercise 4.2
Problem 2.
Prove that $\mathbb{Z} \cong \mathbb{N}$

Proof. Let $f : \mathbb{Z} -> \mathbb{N}$ be given by the function $f(x) = \{\{y = 2x, x \geq 0\}, \{y = 2|x|, x < 0\}\}$
Note that f is injective, i.e. it is one-to-one
Also, f is surjective, i.e. the entire domain of $\mathbb{Z}$ can be found in f and the entire codomain of $\mathbb{N}$ can also be found in f.
So, f is bijective.
However, f is not open since it is closed under the interval $(-\inf, 0) \cup [0, \inf)$
Hence, $\mathbb{Z}$ is not a homeomorphism to $\mathbb{Z}$ as $[0, \inf)$ contains all of its limit points. □

Exercise 4.3
Problem 3.
part i.

Proof. Let X be the unit circle in $\mathbb{R}^2$, i.e. $X = \{< x, y >: x^2 + y^2 = 1\}$ and has the subspace topology.
Note that removing the point $< 1, 0 >$ turns X into an open interval.
So, we have an open interval mapping onto another open interval.
Hence, $X \setminus \{< 1, 0 >\}$ is homeomorphic to the open interval $(0, 1)$

□

1.5. **Chapter 5.** Exercise 5.1
Problem 1.
part i.

Proof. Let $f : (X, \tau) -> (Y, \tau_1)$ be a constant function.
Note that $f(x) = c$, where c is some constant.
So, the open sets under (Y, τ_1) are Y, c, and $\emptyset$.
Also, the open sets under (X, τ) are X, $\mathbb{R}$ and $\emptyset$
For each subset $U \in \tau_1$, there is an inverse $f^{-1}(U) \in \tau$.
Hence, f is continuous. □

part ii.

Proof. Let $f : (X, \tau) -> (X, \tau)$ be the identity function, i.e. $f(x) = x$
Note that the image of the identity function is the pre-image.
So, there are no breaks in the function.
Also, for each subset $U \in \tau$, there is an inverse $f^{-1}(U) \in \tau$.
Hence, f is continuous.

□

Exercise 5.2
Problem 1.

Proof. Since (X, τ) is path-connected, then there exists a continuous function $f : [0, 1] -> (X, \tau)$ such that $f(0) = a$ and $f(1) = b$
Note that $f \circ f$ is a continuous path.
Hence, a continuous image of a path-connected space is path-connected. □

Problem 2.

Proof. Let f be a continuous mapping of the interval $[a, b]$ onto itself, where a and $b \in \mathbb{R}$ and $a < b$.
Note that $f(a) = a$ and $f(b) = b$.
Now, consider a case $f(p)$, where $p \in \mathbb{R}$.
Let $f(a) < f(p) < f(b)$.
So, there is a function $g(x) = x - f(x)$.
Note that g is continuous.
Also, $g(a) = -g(a) < a$ and $g(b) = b - g(b) > a$.
Thus, p is a fixed point between the interval $[a, b]$. □

1.6. **Chapter 6.** Exercise 6.1
Problem 5.

Proof. Let (X, d) be a metric space and $\mathcal{T}$ the corresponding topology on X. Fix $a \in X$.
Let the map $f : (X, \mathcal{T}) -> \mathbb{R}$ be defined by $f(x) = d(a, x)$
Given that d is a metric space, we have the following inequality from definition 6.1.1
$d(a, x) \leq d(a, b) + d(b, x)$ for all $a, b, x \in X$.
By proposition 6.1.23, we have the following:
A subset U of X is open in $(X, \mathcal{T})$ if and only if for each $a \in U$ there exists an $\epsilon > 0$ such that the open ball $B_\epsilon \subseteq U$
Since the metric space is bounded by the triangle inequality and there is an open subset U of X that is open, then the map $f : (X, \mathcal{T}) -> \mathbb{R}$ be defined by $f(x) = d(a, x)$ is continuous. □

Exercise 6.2
Problem 2.

Proof. Let (X, d) be a metric space and $x_1, x_2, ...x_n, ...$ a sequence such that $x_n \to x$ and $x_n \to y$
By definition 6.2.1, x_n converges to x and x_n converges to y.
Since x_n approaches x and x_n approaches y then x and y must be the same value.
Hence, $x = y$.

□

Exercise 6.3
Problem 2.

Proof. Let $\{x_n\} = \{x_1, x_2, x_3, ...x_n, ...\}$ be a Cauchy sequence where $n \in \mathbb{N}$
Let $\{x_a\} = \{x_1, x_2, x_3, ...x_a, ...\}$ be a subsequence of $\{x_n\}$, i.e. $a_1 < a_2 < a_3 < ....$
Note the following from definition 6.3.1
Given any real number $\epsilon > 0$, there exists a positive integer n_0, such that for all integers $m \geq n_0$ and $n \geq n_0$, $d(x_m, x_n) < \epsilon$.
This implies that $\{x_a\}$ follows the same definition for $\{x_n$
Hence, $\{x_a\}$ is indeed a Cauchy sequence.

□

Exercise 6.4
Problem 1.

Proof. Let f be a contraction mapping of the metric space (X, d) and $f : X \to X$.
Definition 6.4.2 gives the following: Let (X, d) be a metric space and f a mapping of X onto itself. Then f is said to be a contraction mapping if there exists an $r \in (0, 1)$ such that $d(f(x_1), f(x_2)) \leq rd(x_1, x_2)$ for all $x_1, x_2 \in X$.
Assume that there is a $\epsilon > 0$.
Let $d(f(x_1), f(x_2)) \leq rd(x_1, x_2) \leq \epsilon$.
Since $rd(x_1), x_2)$ is squeezed between $d(f(x_1), f(x_2))$ and ϵ and r is restricted between 0 and 1 then f is indeed a continuous function. □